

CSCE 628 COMPUTATIONAL BIOLOGY · MODULE 03 · LECTURE NOTES

Scores and Significance

A number is not a finding until it has a null model.

MODULE OUTCOMES

- Explain how substitution matrices are derived.
- Explain why a score needs a null distribution.
- Explain E-values and what they do and do not say.
- Explain the effect of database size on significance.
- Build an empirical null and use it.

1 Substitution matrices

$$s(a,b) = \log [p(a,b) / (q(a) \cdot q(b))]$$

- This is the **log-odds** of observing a aligned with b under a model of relatedness, versus under independence — **positive means more likely than chance, negative means less**, and that is the entire interpretation.
- **PAM matrices are estimated from closely related proteins and then extrapolated** by matrix exponentiation: **PAM250 is the PAM1 model applied 250 times**, which makes it a model of evolution over time — **and the extrapolation is the weak part**, since the model's errors compound.
- **BLOSUM62 is counted directly from blocks of aligned sequences at no more than 62% identity — no extrapolation, counted at the divergence you actually care about — which is why it replaced PAM for most purposes** and is the usual default.
- A substitution score is a **log-odds ratio**, so **adding scores along an alignment multiplies likelihood ratios** — **which is why summing is the right operation** and not merely a convenient one. **The log-odds framing explains both why scores are summed and why some of them must be negative** (§1's callout).

AND THE MATRIX ENCODES AN ASSUMED DIVERGENCE, SO USING THE WRONG ONE COSTS REAL ACCURACY

A matrix tuned for closely related sequences penalises substitutions heavily; one tuned for distant relatives tolerates them — and the right choice depends entirely on what you are looking for, which you must decide before searching.

So using BLOSUM80 to find remote homologues finds fewer of them, and using BLOSUM45 for close matches gives noisier alignments with spurious extensions — both of which are measurable effects rather than matters of preference, and both of which are easy to demonstrate.

Which means the matrix is a stated assumption, and Project 1 asks you to align one pair under two matrices and report the difference, because the difference is frequently larger than expected — sometimes a different alignment entirely, not merely a different score.

And the expected score for a random pair must be negative — otherwise longer alignments always score higher simply by being longer, and local alignment becomes meaningless (Module 02 §2's trap). This is a constraint on how a matrix may be constructed, not an observation.

2 The null model

A SCORE IS ONLY INTERPRETABLE AGAINST THE DISTRIBUTION OF SCORES FROM UNRELATED SEQUENCES

412 is meaningless on its own — the question is how often a pair of unrelated sequences of these lengths and these amino acid compositions would score 412 or better, which is a question about a distribution rather than about this pair.

And that distribution is computable: for ungapped local alignment, the optimal scores follow an extreme value (Gumbel) distribution — which is unsurprising, since an optimal local score is a maximum over many positions — and that is where §3's E-value formula comes from.

Or it is simply measurable: shuffle one sequence a few thousand times, realign each shuffle, and plot the resulting scores — which takes minutes and makes no distributional assumption at all, and which also accounts for composition automatically.

Which is Project 1's central requirement, and the habit this module exists to install — a score without a null is not a result, in this field or any other (CSCE 658 Module 09's permutation testing, CSCE 676 Module 11's significance material).

3 E-values

```
E = K * m * n * exp(-lambda * S)
```

```
S      the alignment score
m, n   query length and DATABASE length
K, lambda constants of the scoring system
```

WHAT IT MEANS

```
the expected NUMBER of alignments scoring at
least S that would occur by chance in a database
of this size.
```

SO

```
E = 10 means expect ten such hits by chance
E = 1e-50 means essentially never by chance
E depends on DATABASE SIZE: the same alignment
      has a worse E-value in a larger database,
      which is correct and surprises people
```

AND NOTE

```
E is not a p-value, though for small E the two
      nearly coincide
E says nothing about whether the hit is
      biologically meaningful (section 4)
```

The same alignment has a worse E-value in a larger database, which is correct **multiple-testing accounting and surprises people every time**: searching more sequences means more chances for a chance match, so the same score is less surprising. **E-values are multiple testing, built into the tool** — which is an unusually good piece of statistical design, and is worth noticing because Module 10 §3 covers the case where nobody built it in for you.

4 Significant and meaningless

- **Low-complexity regions produce spurious significance** — a run of the same residue matches every other such run with a high score — and the null model assumed a composition that the sequence does not have, so the E-value is computed under a false premise.
- **Which is why complexity filtering is on by default** in the standard tools — and why turning it off changes results dramatically, which is worth doing once deliberately to see the effect.
- **And statistical significance is not homology** — convergent evolution and shared short functional motifs both produce genuinely significant matches without any common ancestry — so the inference from one to the other is a judgement.
- **Nor is homology function** — related sequences routinely do different things, particularly after duplication — and **annotation transfer by similarity propagates errors through databases: a mislabelled entry becomes the evidence for the next one**, and the error compounds silently across releases.

- **So the chain is: score → significance → homology → function, and each arrow is an inference somebody has to defend** — which is Module 01 §3's chain in its first concrete instance.

AND THE GENERAL LESSON, WHICH OUTLIVES THE BIOLOGY

Every method in the rest of this course produces a score — an alignment score, a posterior probability, a fold energy, a test statistic — and **every one of them needs the same question asked: what would this look like if there were nothing there?**

Which is the empirical null, and it is almost always available — **permute the labels, shuffle the sequence, randomise the assignment** — and it costs one rerun (Module 01 §4).

And it is the one technique that catches errors in the parts of the pipeline you did not think about, because it tests the whole thing end to end rather than the model you have in your head — a bug that inflates scores inflates the null too, and a bug that leaks labels shows up immediately.

So the habit: build the null before interpreting the score — which is CSCE 676 Module 13's framing and CSCE 658 Module 09's discipline, and is the single most transferable thing in this course.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Henikoff & Henikoff — Amino acid substitution matrices from protein blocks (free)	§1's BLOSUM in the original, with the construction explained.
Karlin & Altschul — Methods for assessing the statistical significance of molecular sequence features (free)	§§2 and 3 — where the E-value formula comes from, and the extreme-value argument.
The NCBI BLAST statistics documentation (free)	§3 — how to read a report, by the people who compute the numbers.
Wootton & Federhen — Statistics of local complexity in amino acid sequences	§4's first caveat — low complexity, and why filtering exists.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Derive the log-odds form** and explain why scores add.
- 2 **Compare BLOSUM45, 62, and 80** on the same distant pair.

- 3 **Check that a matrix's expected random score is negative.**
- 4 **Construct a matrix with all positive entries** and watch local alignment degenerate.
- 5 **Build an empirical null** by shuffling, with 5000 replicates.
- 6 **Place a real score in it** and report the empirical p-value.
- 7 **Compare that to the E-value** the formula gives.
- 8 **Search the same query against two database sizes** and compare E-values.
- 9 **Run one search with and without complexity filtering.**
- 10 **Find one database entry whose annotation was transferred**, and trace it back.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What is a substitution score, formally?
- 2 Why are scores summed?
- 3 Contrast PAM and BLOSUM construction.
- 4 Why must the expected random score be negative?
- 5 Why is a score alone uninterpretable?
- 6 Give two ways to obtain a null distribution.
- 7 Define an E-value, and say what m and n are.
- 8 Why does database size change it, and is that right?
- 9 Give four things a significant hit does not establish.
- 10 State the general lesson and the technique.